

Tensor Products in Quantum Computing

Faisal Alvi

September 19, 2025

1 Introduction

In quantum computing, multi-qubit systems are represented using the mathematical operation known as the *tensor product*, also called the *Kronecker product* in linear algebra. This operation allows us to combine smaller Hilbert spaces into larger ones, enabling the representation of entangled states and multi-qubit gates.

2 Tensor Products: Abstract Definition

Let V and W be vector spaces over a field $\mathbb{F}$ (in quantum computing, $\mathbb{F} = \mathbb{C}$). Let the dimensions of V and W be m and n respectively. The *tensor product* of V and W , denoted $V \otimes W$, creates a *new* vector space of dimension $m \times n$. The elements of $V \otimes W$ are linear combinations of the elements of $V \otimes W$.

In particular, the dimension of $V \otimes W$ is the product of the dimensions of V and W .

$$\dim(V \otimes W) = \dim(V) \cdot \dim(W).$$

In terms of the dirac notation, if $|v\rangle$ is a vector in V , $|w\rangle$ is a vector in W , then $|v\rangle |w\rangle$, $|v, w\rangle$ and $|vw\rangle$ are all used to represent the tensor product $|v\rangle \otimes |w\rangle$.

If $\{v_i\}$ is a basis for V and $\{w_j\}$ is a basis for W , then $\{v_i \otimes w_j\}$ forms a basis for $V \otimes W$.

For the newly created vector space represented by $V \otimes W$, if $|i\rangle$ is a basis for V and $|j\rangle$ is a basis for W , then $|ij\rangle$ is a basis for $V \otimes W$.

3 Kronecker Product as Matrix Representation of the Tensor Product

In the context of quantum computing, the *tensor product* of two vector spaces V and W can be concretely represented using the *Kronecker product* of their matrix (or vector) representations. This allows us to perform explicit calculations with quantum states and operators.

3.1 Definition

Let A be an $m \times n$ matrix and B be a $p \times q$ matrix. i.e.,

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The Kronecker product $A \otimes B$ is the $mp \times nq$ block matrix defined by

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}B & a_{m2}B & \cdots & a_{mn}B \end{pmatrix}.$$

When A and B are column vectors (representing quantum states), this definition reduces to constructing a larger column vector whose entries are products of the entries of A and B .

Example:

Exercises

1. Let $A = I_3$, i.e. identity matrix of dimension 3, and $B = I_2$. What are the elements of the Kronecker (tensor) product $A \otimes B$ and $B \otimes A$? Can some result be concluded from your findings?
2. (HW*) A related term frequently encountered in the study inner products in Quantum Computing is the Kronecker delta i.e. δ_{ij} . Explore what is the Kronecker delta and when is it used?
3. Suppose that $A \otimes B = A$. What can be stated about the matrix B ?

$$\begin{aligned}
& \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \otimes \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} \\
&= \begin{bmatrix} 1 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} & 2 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} \\ 3 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} & 4 \begin{bmatrix} 0 & 5 \\ 6 & 7 \end{bmatrix} \end{bmatrix} \\
&= \left[\begin{array}{cc|cc} 1 \times 0 & 1 \times 5 & 2 \times 0 & 2 \times 5 \\ 1 \times 6 & 1 \times 7 & 2 \times 6 & 2 \times 7 \\ \hline 3 \times 0 & 3 \times 5 & 4 \times 0 & 4 \times 5 \\ 3 \times 6 & 3 \times 7 & 4 \times 6 & 4 \times 7 \end{array} \right] \\
&= \left[\begin{array}{cc|cc} 0 & 5 & 0 & 10 \\ 6 & 7 & 12 & 14 \\ \hline 0 & 15 & 0 & 20 \\ 18 & 21 & 24 & 28 \end{array} \right]
\end{aligned}$$

Figure 1: Example of a Kronecker Product (Wikipedia)

3.2 Kronecker Products of the Basis Vectors

Consider the standard basis states $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

$$\begin{aligned} |0\rangle \otimes |0\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |00\rangle, \\ |0\rangle \otimes |1\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle, \\ |1\rangle \otimes |0\rangle &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |10\rangle, \\ |1\rangle \otimes |1\rangle &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle. \end{aligned}$$

Thus, the Kronecker product naturally builds the computational basis for two-qubit systems.

3.3 Abstract Representation with Coefficients

Suppose we have two arbitrary qubit states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad |\phi\rangle = \gamma|0\rangle + \delta|1\rangle = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}.$$

The tensor (Kronecker) product is

$$\begin{aligned} |\psi\rangle \otimes |\phi\rangle &= \begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix} \\ &= \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle. \end{aligned}$$

3.4 Interpretation in Quantum Computing

This construction emphasizes that the Kronecker product of two single-qubit vectors produces a new vector in the tensor product space $V \otimes W$. In quantum computing terms, it means:

- Two qubits together span a four-dimensional Hilbert space.
- The basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is formed directly from the Kronecker products of the single-qubit basis states.
- More generally, n qubits span a 2^n -dimensional Hilbert space, built recursively via Kronecker products.

In formal terms, each qubit is represented by a vector in the two-dimensional complex Hilbert space

$$\mathcal{H}_2 = \mathbb{C}^2,$$

with computational basis states

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

For two qubits, the joint state space is

$$\mathcal{H}_2 \otimes \mathcal{H}_2 \cong \mathbb{C}^4.$$

The basis states are

$$\{|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle\},$$

which are usually abbreviated as $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. More generally, an n -qubit system lives in

$$\mathcal{H}_2^{\otimes n} = \underbrace{\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2}_{n \text{ times}} \cong \mathbb{C}^{2^n}.$$

4 Basic Properties of the Tensor (Kronecker) Product

In the following discussion A , B and C are matrices. The tensor product has the following important properties:

4.1 Multiplication by a Scalar

Let k be a scalar. Then,

$$k(A \otimes B) = (kA) \otimes B = A \otimes (kB)$$

4.2 Associativity

$$(A \otimes B) \otimes C = A \otimes (B \otimes C).$$

4.3 Distributivity

(Left Distributivity): $A \otimes (B + C) = A \otimes B + A \otimes C$.

(Right Distributivity): $(A + B) \otimes C = A \otimes C + B \otimes C$.

4.4 Non-commutativity

In general, (but compare with Exercise 1 on Page 2)

$$A \otimes B \neq B \otimes A.$$

4.5 Transpose Property

$$(A \otimes B)^T = A^T \otimes B^T$$

4.6 Conjugate Property

$$(A \otimes B)^* = A^* \otimes B^*$$

4.7 Conjugate Transpose Property

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$$

4.8 Inverse Property

If A and B are invertible, then

$$(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$$

4.9 Mixed-product Property

If A, B, C and D are compatible matrices, such that the matrix products AC and BD exist, then

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD).$$

4.10 ‘Exponentiation’ or ‘Power’ for tensor products

Sometimes the notation $\otimes^n$ is used to represent the repeated tensor product of an operator (or the repeated kronecker product of a matrix) with itself.

For example, let $|\psi\rangle$ be a vector. Then a tensor product of $|\psi\rangle$ with itself 3 times will be represented as $|\psi\rangle^{\otimes 3}$.

Exercises

Taken from Nielsen and Chuang (the book is available online as well as on our Canvas site).

4. Exercise 2.26 (page 74)
5. (HW*) Exercise 2.27 (page 74)
6. Exercise 2.29 (page 74)

For reference, the snapshot of the book for this page is shown below:

Exercise 2.26: Let $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Write out $|\psi\rangle^{\otimes 2}$ and $|\psi\rangle^{\otimes 3}$ explicitly, both in terms of tensor products like $|0\rangle|1\rangle$, and using the Kronecker product.

Exercise 2.27: Calculate the matrix representation of the tensor products of the Pauli operators (a) X and Z ; (b) I and X ; (c) X and I . Is the tensor product commutative?

Exercise 2.28: Show that the transpose, complex conjugation, and adjoint operations distribute over the tensor product,

$$(A \otimes B)^* = A^* \otimes B^*; (A \otimes B)^T = A^T \otimes B^T; (A \otimes B)^\dagger = A^\dagger \otimes B^\dagger. (2.53)$$

Exercise 2.29: Show that the tensor product of two unitary operators is unitary.

Figure 2: Snapshot of Nielsen and Chuang’s Book

5 Tensor Products and Circuit Diagrams

So far, we have seen tensor (Kronecker) products in an abstract and matrix setting. Let us now connect them to quantum circuits.

The n -qubit initial state of a quantum circuit is given by

$$|0\rangle^{\otimes n} = |0\rangle \otimes |0\rangle \otimes \cdots \otimes |0\rangle.$$

This expression means that all n lines (or wires) in the circuit start in the $|0\rangle$ state. For example, for three qubits:

$$|0\rangle^{\otimes 3} = |0\rangle \otimes |0\rangle \otimes |0\rangle = |000\rangle.$$

$$\begin{array}{c} |0\rangle \text{ ---} \\ |0\rangle \text{ ---} \\ |0\rangle \text{ ---} \end{array}$$

Here, the three lines (or wires) initialized to $|0\rangle$ represent the state $|000\rangle$.

6 Operators on Multiple lines (or wires)

If we apply operators to different lines (or wires) independently, the overall operator is described by the tensor product of those operators.

For instance, applying U on the first qubit and V on the second qubit means that the two-qubit operator is

$$U \otimes V.$$

Example

Consider $X \otimes I$ acting on the two-qubit state $|01\rangle$:

$$(X \otimes I)(|0\rangle \otimes |1\rangle) = (X|0\rangle) \otimes (I|1\rangle) = |1\rangle \otimes |1\rangle = |11\rangle.$$

Circuit representation:

$$\begin{array}{c} |0\rangle \text{ ---} \boxed{X} \text{ ---} \\ |1\rangle \text{ ---} \boxed{I} \text{ ---} \end{array}$$

This shows that the operator $X \otimes I$ flips the first qubit while leaving the second unchanged. Note, how the mixed-product property is being used.

7 Mixed-Product Property in Circuits

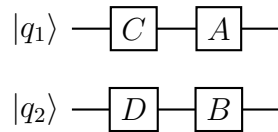
Recall the mixed-product property:

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD).$$

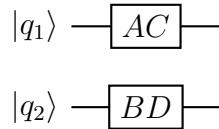
Circuit interpretation

- First apply C on qubit 1 and D on qubit 2.
- Then apply A on qubit 1 and B on qubit 2.

This is equivalent to applying AC on the first qubit and BD on the second qubit.



is equivalent to



This diagrammatic view makes the mixed-product property concrete: independent operators on different lines (or wires) combine exactly as the tensor product rule states, i.e. $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.

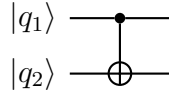
8 Beyond Independent lines (or wires)

Not all operations on multiple qubits can be written as tensor products. Some gates, such as the CNOT or controlled- Z , *entangle* qubits and cannot be factored into $U \otimes V$ form.

For instance, the CNOT gate on two qubits has the matrix

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix},$$

which is not separable into $A \otimes B$.



This distinction is central: **tensor product operators represent independent (single-qubit) operations, while multi-qubit gates like CNOT go beyond tensor products, creating entanglement.**

Exercises

7. (HW*) Apply $H \otimes H$ to the state $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ where

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Expand the resulting state fully in terms of computational basis states $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

(Hint: $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.)

What kind of superposition do you obtain?

8. (Challenge) Attempt only if you are ambitious! (Exercise 2.33 - Nielsen and Chuang)

Exercise 2.33: The Hadamard operator on one qubit may be written as

$$H = \frac{1}{\sqrt{2}} \left[(|0\rangle + |1\rangle)\langle 0| + (|0\rangle - |1\rangle)\langle 1| \right]. \quad (2.54)$$

Show explicitly that the Hadamard transform on n qubits, $H^{\otimes n}$, may be written as

$$H^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x,y} (-1)^{x \cdot y} |x\rangle \langle y|. \quad (2.55)$$

Write out an explicit matrix representation for $H^{\otimes 2}$.

Figure 3: Challenge Question